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### Object Interaction During A Contact Center Engagement Video Conference

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#### Abstract

Interactions with objects depicted within video streams displayed during a video conference are detected to cause information associated with the interacted objects to be presented. During a video conference, multiple video layers of a video stream obtained from a first participant device connected to the video conference are identified. An interaction with an object within one of those multiple video layers is detected during the video conference, in which the interaction is from a second participant device connected to the video conference. Based on the interaction, information associated with the object is presented during the video conference within a graphical user interface associated with the video conference. The video stream may, for example, initially include a background layer, a foreground layer, and an overlay layer. Interactive video layers corresponding to each of those initial layers may be introduced to receive interactions with objects depicted therein.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation of U.S. application Ser. No. 18/296,728, filed Apr. 6, 2023, which is a continuation of U.S. application Ser. No. 17/976,288, filed Oct. 28, 2022, the entire disclosures of which are herein incorporated by reference.

### FIELD

[0002] This disclosure generally relates to video conferencing, and, more specifically, to enabling interactions with objects within video layers of video streams displayed during a video conference.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0003] This disclosure is best understood from the following detailed description when read in conjunction with the accompanying drawings. It is emphasized that, according to common practice, the various features of the drawings are not to-scale. On the contrary, the dimensions of the various features are arbitrarily expanded or reduced for clarity.

[0004] FIG. 1 is a block diagram of an example of an electronic computing and communications system.

[0005] FIG. 2 is a block diagram of an example internal configuration of a computing device of an electronic computing and communications system.

[0006] FIG. 3 is a block diagram of an example of a software platform implemented by an electronic computing and communications system.

[0007] FIG. 4 is a block diagram of an example of a conferencing system for delivering conferencing software services in an electronic computing and communications system.

[0008] FIG. 5 is a block diagram of an example of an interactive video layer system.

[0009] FIG. 6 is a block diagram of an example of functionality of interactive video layer software.

[0010] FIG. 7 is an illustration of an example of video layers of a video stream output for display during a video conference.

[0011] FIGS. 8A-B are illustrations of example graphical user interfaces (GUIs) of a participant-to-participant video conference facilitated using interactive video layers.

[0012] FIGS. 9A-B are illustrations of example GUIs of a contact center engagement video conference facilitated using interactive video layers.

[0013] FIGS. 10A-B are illustrations of example GUIs of an online learning video conference facilitated using interactive video layers.

[0014] FIG. 11 is a flowchart of an example of a technique for enabling and processing interactions with objects within video layers of a video conference.

## DETAILED DESCRIPTION

[0015] Conferencing software is frequently used across various industries to support video-enabled conferences between participants in multiple locations. In some cases, each of the conference participants separately connects to the conferencing software from their own remote locations. In other cases, one or more of the conference participants may be physically located in and connect to the conferencing software from a conference room or similar physical space (e.g., in an office setting) while other conference participants connect to the conferencing software from one or more remote locations. Conferencing software thus enables people to conduct video conferences without requiring them to be physically present with one another. Conferencing software may be available as a standalone software product or it may be integrated within a software platform, such as a unified communications as a service (UCaaS) platform.

[0016] The conferencing software implementing a video conference obtains video streams from devices of conference participants, processes the video streams for formatting and bandwidth control, and transmits the processed video streams to the various participant devices for display thereat. Generally, a video stream obtained from a device of a conference participant and displayed during a video conference depicts the conference participant, so that the many participants can see and therefore more personally interact with one another despite being remote from one another. However, a video stream may depict content beyond or other than the conference participant. For example, a video stream may depict one or more objects behind, next to, or otherwise nearby to the subject conference participant. Examples of such objects include, but are not limited to, decorations (e.g., art, diplomas, or tchotchkes), furniture (e.g., chairs, desks, or bookshelves), and personal items (e.g., books, family pictures, or electronic devices).

[0017] A participant of a video conference may wish to learn more about the contents depicted within a video stream of another participant during the video conference. For example, an object appearing behind a participant within a video stream may be something another participant is interested in purchasing. In another example, a participant may simply be curious about that object and want to know more about it, independent of any interest in purchasing it. Because conventional video conferencing approaches do not enable interactions with participant video streams, the interested participant (i.e., the participant interested in the object) must communicate with the participant whose video stream includes the object to ask about the object, either by asking about the object out loud during the conference or by sending a message within a chat or like window of the video conference. However, both of these approaches suffer drawbacks. For example, speech may disrupt a discussion during the video conference, especially where the discussion does not involve the interested participant, or may be misheard and require repeating, thereby taking more time away from the subject of the video conference. In another example, chat or other messages may not be seen by the participant to whom they are directed until much later during the video conference, thereby either delaying or entirely precluding a response. In yet another example, with speech or messaging, even where the party to whom the speech or message is directed timely perceives the speech or message, it is possible that they will not know enough about the object to satisfy the interested party.

[0018] Implementations of this disclosure address problems such as these using an interactive video layer system which enables interactions with objects within video layers of video streams displayed during a video conference. During a video conference, multiple video layers of a video stream obtained from a first participant device connected to the video conference are identified. A video stream obtained from a participant device (i.e., a device of a participant of the video conference) may be considered to include one or more video layers, for example, a background layer (e.g., a virtual background or a portion of a video stream showing an actual background), a foreground layer (e.g., showing the human participant(s) and any objects held thereby; sometimes referred as the human layer), and an overlay layer (e.g., showing participant, company, or information, such as in text and/or images). For each initial video layer of the video stream, the

interactive video layer system generates an interactive video layer which enables interactions with objects within the corresponding video layer. For example, a first interactive video layer may correspond to the background layer and enable interactions with objects shown within the video background, while a second interactive video layer may correspond to a foreground layer and enable interactions with objects shown within the video foreground.

[0019] An interaction with an object within one of those multiple video layers is detected during the video conference, in which the interaction is from a second participant device connected to the video conference. Based on the interaction, information associated with the object is presented during the video conference within a GUI associated with the video conference. That is, interactions within an object may trigger one or more events to occur, such as the presentation of a pop-up window including embedded information from an external source (e.g., an online encyclopedia article or online marketplace listing) or a hyperlink to such an external source, a graphic showing information associated with the object or with an event corresponding to the interaction with the object (e.g., poll results showing votes of objects within a participant video stream based on numbers of interactions therewith), or an option to share information associated with the interacted object over a modality separate from the video conference (e.g., another communication approach enabled by a UCaaS platform or other platform which implements the video conference). The types of interactive objects, interactive video layers, and degrees of interaction (e.g., what happens when something is interacted with) may be configured by a host of the video conference or another participant thereof.

[0020] The implementations of this disclosure enable interactions with objects within video layers during many different video conference settings. In one example, interactive video layers may be used with a participant-to-participant video conference (e.g., a conventional video conference between two or more given people), in which participants may have their own user interface tiles and video streams, connect from their own participant devices, have similar meeting controls, be or be granted host privileges, or the like. In another example, interactive video layers may be used with a contact center engagement video conference in which one participant to the video conference is a contact center user who is accessing a contact center over a video modality and another participant to the video conference is a contact center agent who works with or otherwise for the contact center to address queries from contact center users. In yet another example, interactive video layers may be used with an online learning video conference in which multiple participants are recognized generally as audience participants and one or more select participants are recognized generally as leader participants (e.g., for educational, webinar, or other purposes).

[0021] To describe some implementations in greater detail, reference is first made to examples of hardware and software structures used to implement an interactive video layer system. FIG. 1 is a block diagram of an example of an electronic computing and communications system **100**, which can be or include a distributed computing system (e.g., a client-server computing system), a cloud computing system, a clustered computing system, or the like.

[0022] The system **100** includes one or more customers, such as customers **102A** through **102B**, which may each be a public entity, private entity, or another corporate entity or individual that purchases or otherwise uses software services, such as of a UCaaS platform provider. Each customer can include one or more clients. For example, as shown and without limitation, the customer **102A** can include clients **104A** through **104B**, and the customer **102B** can include clients **104C** through **104D**. A customer can include a customer network or domain. For example, and without limitation, the clients **104A** through **104B** can be associated or communicate with a customer network or domain for the customer **102A** and the clients **104C** through **104D** can be associated or communicate with a customer network or domain for the customer **102B**.

[0023] A client, such as one of the clients **104A** through **104D**, may be or otherwise refer to one or both of a client device or a client application. Where a client is or refers to a client device, the client can comprise a computing system, which can include one or more computing devices, such as a

mobile phone, a tablet computer, a laptop computer, a notebook computer, a desktop computer, or another suitable computing device or combination of computing devices. Where a client instead is or refers to a client application, the client can be an instance of software running on a customer device (e.g., a client device or another device). In some implementations, a client can be implemented as a single physical unit or as a combination of physical units. In some implementations, a single physical unit can include multiple clients.

[0024] The system **100** can include a number of customers and/or clients or can have a configuration of customers or clients different from that generally illustrated in FIG. **1**. For example, and without limitation, the system **100** can include hundreds or thousands of customers, and at least some of the customers can include or be associated with a number of clients.

[0025] The system **100** includes a datacenter **106**, which may include one or more servers. The datacenter **106** can represent a geographic location, which can include a facility, where the one or more servers are located. The system **100** can include a number of datacenters and servers or can include a configuration of datacenters and servers different from that generally illustrated in FIG. **1**. For example, and without limitation, the system **100** can include tens of datacenters, and at least some of the datacenters can include hundreds or another suitable number of servers. In some implementations, the datacenter **106** can be associated or communicate with one or more datacenter networks or domains, which can include domains other than the customer domains for the customers **102A** through **102B**.

[0026] The datacenter **106** includes servers used for implementing software services of a UCaaS platform. The datacenter **106** as generally illustrated includes an application server **108**, a database server **110**, and a telephony server **112**. The servers **108** through **112** can each be a computing system, which can include one or more computing devices, such as a desktop computer, a server computer, or another computer capable of operating as a server, or a combination thereof. A suitable number of each of the servers **108** through **112** can be implemented at the datacenter **106**. The UCaaS platform uses a multi-tenant architecture in which installations or instantiations of the servers **108** through **112** is shared amongst the customers **102A** through **102B**.

[0027] In some implementations, one or more of the servers **108** through **112** can be a non-hardware server implemented on a physical device, such as a hardware server. In some implementations, a combination of two or more of the application server **108**, the database server **110**, and the telephony server **112** can be implemented as a single hardware server or as a single non-hardware server implemented on a single hardware server. In some implementations, the datacenter **106** can include servers other than or in addition to the servers **108** through **112**, for example, a media server, a proxy server, or a web server.

[0028] The application server **108** runs web-based software services deliverable to a client, such as one of the clients **104A** through **104D**. As described above, the software services may be of a UCaaS platform. For example, the application server **108** can implement all or a portion of a UCaaS platform, including conferencing software, messaging software, and/or other intra-party or inter-party communications software. The application server **108** may, for example, be or include a unitary Java Virtual Machine (JVM).

[0029] In some implementations, the application server **108** can include an application node, which can be a process executed on the application server **108**. For example, and without limitation, the application node can be executed in order to deliver software services to a client, such as one of the clients **104A** through **104D**, as part of a software application. The application node can be implemented using processing threads, virtual machine instantiations, or other computing features of the application server **108**. In some such implementations, the application server **108** can include a suitable number of application nodes, depending upon a system load or other characteristics associated with the application server **108**. For example, and without limitation, the application server **108** can include two or more nodes forming a node cluster. In some such implementations, the application nodes implemented on a single application server **108** can run on different hardware

servers.

[0030] The database server **110** stores, manages, or otherwise provides data for delivering software services of the application server **108** to a client, such as one of the clients **104A** through **104D**. In particular, the database server **110** may implement one or more databases, tables, or other information sources suitable for use with a software application implemented using the application server **108**. The database server **110** may include a data storage unit accessible by software executed on the application server **108**. A database implemented by the database server **110** may be a relational database management system (RDBMS), an object database, an XML database, a configuration management database (CMDB), a management information base (MIB), one or more flat files, other suitable non-transient storage mechanisms, or a combination thereof. The system **100** can include one or more database servers, in which each database server can include one, two, three, or another suitable number of databases configured as or comprising a suitable database type or combination thereof.

[0031] In some implementations, one or more databases, tables, other suitable information sources, or portions or combinations thereof may be stored, managed, or otherwise provided by one or more of the elements of the system **100** other than the database server **110**, for example, the client **104** or the application server **108**.

[0032] The telephony server **112** enables network-based telephony and web communications from and to clients of a customer, such as the clients **104A** through **104B** for the customer **102A** or the clients **104C** through **104D** for the customer **102B**. Some or all of the clients **104A** through **104D** may be voice over internet protocol (VOIP)-enabled devices configured to send and receive calls over a network **114**. In particular, the telephony server **112** includes a session initiation protocol (SIP) zone and a web zone. The SIP zone enables a client of a customer, such as the customer **102A** or **102B**, to send and receive calls over the network **114** using SIP requests and responses. The web zone integrates telephony data with the application server **108** to enable telephony-based traffic access to software services run by the application server **108**. Given the combined functionality of the SIP zone and the web zone, the telephony server **112** may be or include a cloud-based private branch exchange (PBX) system.

[0033] The SIP zone receives telephony traffic from a client of a customer and directs same to a destination device. The SIP zone may include one or more call switches for routing the telephony traffic. For example, to route a VOIP call from a first VOIP-enabled client of a customer to a second VOIP-enabled client of the same customer, the telephony server **112** may initiate a SIP transaction between a first client and the second client using a PBX for the customer. However, in another example, to route a VOIP call from a VOIP-enabled client of a customer to a client or non-client device (e.g., a desktop phone which is not configured for VOIP communication) which is not VOIP-enabled, the telephony server **112** may initiate a SIP transaction via a VOIP gateway that transmits the SIP signal to a public switched telephone network (PSTN) system for outbound communication to the non-VOIP-enabled client or non-client phone. Hence, the telephony server **112** may include a PSTN system and may in some cases access an external PSTN system.

[0034] The telephony server **112** includes one or more session border controllers (SBCs) for interfacing the SIP zone with one or more aspects external to the telephony server **112**. In particular, an SBC can act as an intermediary to transmit and receive SIP requests and responses between clients or non-client devices of a given customer with clients or non-client devices external to that customer. When incoming telephony traffic for delivery to a client of a customer, such as one of the clients **104A** through **104D**, originating from outside the telephony server **112** is received, a SBC receives the traffic and forwards it to a call switch for routing to the client.

[0035] In some implementations, the telephony server **112**, via the SIP zone, may enable one or more forms of peering to a carrier or customer premise. For example, Internet peering to a customer premise may be enabled to ease the migration of the customer from a legacy provider to a service provider operating the telephony server **112**. In another example, private peering to a

customer premise may be enabled to leverage a private connection terminating at one end at the telephony server **112** and at the other end at a computing aspect of the customer environment. In yet another example, carrier peering may be enabled to leverage a connection of a peered carrier to the telephony server **112**.

[0036] In some such implementations, a SBC or telephony gateway within the customer environment may operate as an intermediary between the SBC of the telephony server **112** and a PSTN for a peered carrier. When an external SBC is first registered with the telephony server **112**, a call from a client can be routed through the SBC to a load balancer of the SIP zone, which directs the traffic to a call switch of the telephony server **112**. Thereafter, the SBC may be configured to communicate directly with the call switch.

[0037] The web zone receives telephony traffic from a client of a customer, via the SIP zone, and directs same to the application server **108** via one or more Domain Name System (DNS) resolutions. For example, a first DNS within the web zone may process a request received via the SIP zone and then deliver the processed request to a web service which connects to a second DNS at or otherwise associated with the application server **108**. Once the second DNS resolves the request, it is delivered to the destination service at the application server **108**. The web zone may also include a database for authenticating access to a software application for telephony traffic processed within the SIP zone, for example, a softphone.

[0038] The clients **104A** through **104D** communicate with the servers **108** through **112** of the datacenter **106** via the network **114**. The network **114** can be or include, for example, the Internet, a local area network (LAN), a wide area network (WAN), a virtual private network (VPN), or another public or private means of electronic computer communication capable of transferring data between a client and one or more servers. In some implementations, a client can connect to the network **114** via a communal connection point, link, or path, or using a distinct connection point, link, or path. For example, a connection point, link, or path can be wired, wireless, use other communications technologies, or a combination thereof.

[0039] The network **114**, the datacenter **106**, or another element, or combination of elements, of the system **100** can include network hardware such as routers, switches, other network devices, or combinations thereof. For example, the datacenter **106** can include a load balancer **116** for routing traffic from the network **114** to various servers associated with the datacenter **106**. The load balancer **116** can route, or direct, computing communications traffic, such as signals or messages, to respective elements of the datacenter **106**.

[0040] For example, the load balancer **116** can operate as a proxy, or reverse proxy, for a service, such as a service provided to one or more remote clients, such as one or more of the clients **104A** through **104D**, by the application server **108**, the telephony server **112**, and/or another server. Routing functions of the load balancer **116** can be configured directly or via a DNS. The load balancer **116** can coordinate requests from remote clients and can simplify client access by masking the internal configuration of the datacenter **106** from the remote clients.

[0041] In some implementations, the load balancer **116** can operate as a firewall, allowing or preventing communications based on configuration settings. Although the load balancer **116** is depicted in FIG. **1** as being within the datacenter **106**, in some implementations, the load balancer **116** can instead be located outside of the datacenter **106**, for example, when providing global routing for multiple datacenters. In some implementations, load balancers can be included both within and outside of the datacenter **106**. In some implementations, the load balancer **116** can be omitted.

[0042] FIG. **2** is a block diagram of an example internal configuration of a computing device **200** of an electronic computing and communications system. In one configuration, the computing device **200** may implement one or more of the client **104**, the application server **108**, the database server **110**, or the telephony server **112** of the system **100** shown in FIG. **1**.

[0043] The computing device **200** includes components or units, such as a processor **202**, a

memory **204**, a bus **206**, a power source **208**, peripherals **210**, a user interface **212**, a network interface **214**, other suitable components, or a combination thereof. One or more of the memory **204**, the power source **208**, the peripherals **210**, the user interface **212**, or the network interface **214** can communicate with the processor **202** via the bus **206**.

[0044] The processor **202** is a central processing unit, such as a microprocessor, and can include single or multiple processors having single or multiple processing cores. Alternatively, the processor **202** can include another type of device, or multiple devices, configured for manipulating or processing information. For example, the processor **202** can include multiple processors interconnected in one or more manners, including hardwired or networked. The operations of the processor **202** can be distributed across multiple devices or units that can be coupled directly or across a local area or other suitable type of network. The processor **202** can include a cache, or cache memory, for local storage of operating data or instructions.

[0045] The memory **204** includes one or more memory components, which may each be volatile memory or non-volatile memory. For example, the volatile memory can be random access memory (RAM) (e.g., a DRAM module, such as DDR SDRAM). In another example, the non-volatile memory of the memory **204** can be a disk drive, a solid state drive, flash memory, or phase-change memory. In some implementations, the memory **204** can be distributed across multiple devices. For example, the memory **204** can include network-based memory or memory in multiple clients or servers performing the operations of those multiple devices.

[0046] The memory **204** can include data for immediate access by the processor **202**. For example, the memory **204** can include executable instructions **216**, application data **218**, and an operating system **220**. The executable instructions **216** can include one or more application programs, which can be loaded or copied, in whole or in part, from non-volatile memory to volatile memory to be executed by the processor **202**. For example, the executable instructions **216** can include instructions for performing some or all of the techniques of this disclosure. The application data **218** can include user data, database data (e.g., database catalogs or dictionaries), or the like. In some implementations, the application data **218** can include functional programs, such as a web browser, a web server, a database server, another program, or a combination thereof. The operating system **220** can be, for example, Microsoft Windows®, Mac OS X®, or Linux®; an operating system for a mobile device, such as a smartphone or tablet device; or an operating system for a non-mobile device, such as a mainframe computer.

[0047] The power source **208** provides power to the computing device **200**. For example, the power source **208** can be an interface to an external power distribution system. In another example, the power source **208** can be a battery, such as where the computing device **200** is a mobile device or is otherwise configured to operate independently of an external power distribution system. In some implementations, the computing device **200** may include or otherwise use multiple power sources. In some such implementations, the power source **208** can be a backup battery.

[0048] The peripherals **210** includes one or more sensors, detectors, or other devices configured for monitoring the computing device **200** or the environment around the computing device **200**. For example, the peripherals **210** can include a geolocation component, such as a global positioning system location unit. In another example, the peripherals can include a temperature sensor for measuring temperatures of components of the computing device **200**, such as the processor **202**. In some implementations, the computing device **200** can omit the peripherals **210**.

[0049] The user interface **212** includes one or more input interfaces and/or output interfaces. An input interface may, for example, be a positional input device, such as a mouse, touchpad, touchscreen, or the like; a keyboard; or another suitable human or machine interface device. An output interface may, for example, be a display, such as a liquid crystal display, a cathode-ray tube, a light emitting diode display, or other suitable display.

[0050] The network interface **214** provides a connection or link to a network (e.g., the network **114** shown in FIG. 1). The network interface **214** can be a wired network interface or a wireless

network interface. The computing device **200** can communicate with other devices via the network interface **214** using one or more network protocols, such as using Ethernet, transmission control protocol (TCP), internet protocol (IP), power line communication, an IEEE 802.X protocol (e.g., Wi-Fi, Bluetooth, or ZigBee), infrared, visible light, general packet radio service (GPRS), global system for mobile communications (GSM), code-division multiple access (CDMA), Z-Wave, another protocol, or a combination thereof.

[0051] FIG. **3** is a block diagram of an example of a software platform **300** implemented by an electronic computing and communications system, for example, the system **100** shown in FIG. **1**. The software platform **300** is a UCaaS platform accessible by clients of a customer of a UCaaS platform provider, for example, the clients **104A** through **104B** of the customer **102A** or the clients **104C** through **104D** of the customer **102B** shown in FIG. **1**. The software platform **300** may be a multi-tenant platform instantiated using one or more servers at one or more datacenters including, for example, the application server **108**, the database server **110**, and the telephony server **112** of the datacenter **106** shown in FIG. **1**.

[0052] The software platform **300** includes software services accessible using one or more clients. For example, a customer **302** as shown includes four clients **304** through **310**—a desk phone, a computer, a mobile device, and a shared device. The desk phone is a desktop unit configured to at least send and receive calls and includes an input device for receiving a telephone number or extension to dial to and an output device for outputting audio and/or video for a call in progress. The computer is a desktop, laptop, or tablet computer including an input device for receiving some form of user input and an output device for outputting information in an audio and/or visual format. The mobile device is a smartphone, wearable device, or other mobile computing aspect including an input device for receiving some form of user input and an output device for outputting information in an audio and/or visual format. The desk phone, the computer, and the mobile device may generally be considered personal devices configured for use by a single user. The shared device is a desk phone, a computer, a mobile device, or a different device which may instead be configured for use by multiple specified or unspecified users.

[0053] Each of the clients **304** through **310** includes or runs on a computing device configured to access at least a portion of the software platform **300**. In some implementations, the customer **302** may include additional clients not shown. For example, the customer **302** may include multiple clients of one or more client types (e.g., multiple desk phones or multiple computers) and/or one or more clients of a client type not shown in FIG. **3** (e.g., wearable devices or televisions other than as shared devices). For example, the customer **302** may have tens or hundreds of desk phones, computers, mobile devices, and/or shared devices.

[0054] The software services of the software platform **300** generally relate to communications tools, but are in no way limited in scope. As shown, the software services of the software platform **300** include telephony software **312**, conferencing software **314**, messaging software **316**, and other software **318**. Some or all of the software **312** through **318** uses customer configurations **320** specific to the customer **302**. The customer configurations **320** may, for example, be data stored within a database or other data store at a database server, such as the database server **110** shown in FIG. **1**.

[0055] The telephony software **312** enables telephony traffic between ones of the clients **304** through **310** and other telephony-enabled devices, which may be other ones of the clients **304** through **310**, other VOIP-enabled clients of the customer **302**, non-VOIP-enabled devices of the customer **302**, VOIP-enabled clients of another customer, non-VOIP-enabled devices of another customer, or other VOIP-enabled clients or non-VOIP-enabled devices. Calls sent or received using the telephony software **312** may, for example, be sent or received using the desk phone, a softphone running on the computer, a mobile application running on the mobile device, or using the shared device that includes telephony features.

[0056] The telephony software **312** further enables phones that do not include a client application

to connect to other software services of the software platform **300**. For example, the telephony software **312** may receive and process calls from phones not associated with the customer **302** to route that telephony traffic to one or more of the conferencing software **314**, the messaging software **316**, or the other software **318**.

[0057] The conferencing software **314** enables audio, video, and/or other forms of conferences between multiple participants, such as to facilitate a conference between those participants. In some cases, the participants may all be physically present within a single location, for example, a conference room, in which the conferencing software **314** may facilitate a conference between only those participants and using one or more clients within the conference room. In some cases, one or more participants may be physically present within a single location and one or more other participants may be remote, in which the conferencing software **314** may facilitate a conference between all of those participants using one or more clients within the conference room and one or more remote clients. In some cases, the participants may all be remote, in which the conferencing software **314** may facilitate a conference between the participants using different clients for the participants. The conferencing software **314** can include functionality for hosting, presenting scheduling, joining, or otherwise participating in a conference. The conferencing software **314** may further include functionality for recording some or all of a conference and/or documenting a transcript for the conference.

[0058] The messaging software **316** enables instant messaging, unified messaging, and other types of messaging communications between multiple devices, such as to facilitate a chat or other virtual conversation between users of those devices. The unified messaging functionality of the messaging software **316** may, for example, refer to email messaging which includes a voicemail transcription service delivered in email format.

[0059] The other software **318** enables other functionality of the software platform **300**. Examples of the other software **318** include, but are not limited to, device management software, resource provisioning and deployment software, administrative software, third party integration software, and the like. In one particular example, the other software **318** can include interactive video layer software for enabling interactions with objects within video layers of video streams displayed during a video conference. In some such cases, the other software **318** may be or be included in the conferencing software **314**.

[0060] The software **312** through **318** may be implemented using one or more servers, for example, of a datacenter such as the datacenter **106** shown in FIG. **1**. For example, one or more of the software **312** through **318** may be implemented using an application server, a database server, and/or a telephony server, such as the servers **108** through **112** shown in FIG. **1**. In another example, one or more of the software **312** through **318** may be implemented using servers not shown in FIG. **1**, for example, a meeting server, a web server, or another server. In yet another example, one or more of the software **312** through **318** may be implemented using one or more of the servers **108** through **112** and one or more other servers. The software **312** through **318** may be implemented by different servers or by the same server.

[0061] Features of the software services of the software platform **300** may be integrated with one another to provide a unified experience for users. For example, the messaging software **316** may include a user interface element configured to initiate a call with another user of the customer **302**. In another example, the telephony software **312** may include functionality for elevating a telephone call to a conference. In yet another example, the conferencing software **314** may include functionality for sending and receiving instant messages between participants and/or other users of the customer **302**. In yet another example, the conferencing software **314** may include functionality for file sharing between participants and/or other users of the customer **302**. In some implementations, some or all of the software **312** through **318** may be combined into a single software application run on clients of the customer, such as one or more of the clients **304** through **310**.

[0062] FIG. 4 is a block diagram of an example of a conferencing system 400 for delivering conferencing software services in an electronic computing and communications system, for example, the system 100 shown in FIG. 1. The conferencing system 400 includes a thread encoding tool 402, a switching/routing tool 404, and conferencing software 406. The conferencing software 406, which may, for example, be the conferencing software 314 shown in FIG. 3, is software for implementing conferences (e.g., video conferences) between users of clients and/or phones, such as clients 408 and 410 and phone 412. For example, the clients 408 or 410 may each be one of the clients 304 through 310 shown in FIG. 3 that runs a client application associated with the conferencing software 406, and the phone 412 may be a telephone which does not run a client application associated with the conferencing software 406 or otherwise access a web application associated with the conferencing software 406. The conferencing system 400 may in at least some cases be implemented using one or more servers of the system 100, for example, the application server 108 shown in FIG. 1. Although two clients and a phone are shown in FIG. 4, other numbers of clients and/or other numbers of phones can connect to the conferencing system 400.

[0063] Implementing a conference includes transmitting and receiving video, audio, and/or other data between clients and/or phones, as applicable, of the conference participants. Each of the client 408, the client 410, and the phone 412 may connect through the conferencing system 400 using separate input streams to enable users thereof to participate in a conference together using the conferencing software 406. The various channels used for establishing connections between the clients 408 and 410 and the phone 412 may, for example, be based on the individual device capabilities of the clients 408 and 410 and the phone 412.

[0064] The conferencing software 406 includes a user interface tile for each input stream received and processed at the conferencing system 400. A user interface tile as used herein generally refers to a portion of a conferencing software user interface which displays information (e.g., a rendered video) associated with one or more conference participants. A user interface tile may, but need not, be generally rectangular. The size of a user interface tile may depend on one or more factors including the view style set for the conferencing software user interface at a given time and whether the one or more conference participants represented by the user interface tile are active speakers at a given time. The view style for the conferencing software user interface, which may be uniformly configured for all conference participants by a host of the subject conference or which may be individually configured by each conference participant, may be one of a gallery view in which all user interface tiles are similarly or identically sized and arranged in a generally grid layout or a speaker view in which one or more user interface tiles for active speakers are enlarged and arranged in a center position of the conferencing software user interface while the user interface tiles for other conference participants are reduced in size and arranged near an edge of the conferencing software user interface. In some cases, the view style or one or more other configurations related to the display of user interface tiles may be based on a type of video conference implemented using the conferencing software 406 (e.g., a participant-to-participant video conference, a contact center engagement video conference, or an online learning video conference, as will be described below).

[0065] The content of the user interface tile associated with a given participant may be dependent upon the source of the input stream for that participant. For example, where a participant accesses the conferencing software 406 from a client, such as the client 408 or 410, the user interface tile associated with that participant may include a video stream captured at the client and transmitted to the conferencing system 400, which is then transmitted from the conferencing system 400 to other clients for viewing by other participants (although the participant may optionally disable video features to suspend the video stream from being presented during some or all of the conference). In another example, where a participant accesses the conferencing software 406 from a phone, such as the phone 412, the user interface tile for the participant may be limited to a static image showing text (e.g., a name, telephone number, or other identifier associated with the participant or the phone

**412**) or other default background aspect since there is no video stream presented for that participant.

[0066] The thread encoding tool **402** receives video streams separately from the clients **408** and **410** and encodes those video streams using one or more transcoding tools, such as to produce variant streams at different resolutions. For example, a given video stream received from a client may be processed using multi-stream capabilities of the conferencing system **400** to result in multiple resolution versions of that video stream, including versions at 90 p, 180 p, 360 p, 720 p, and/or 1080 p, amongst others. The video streams may be received from the clients over a network, for example, the network **114** shown in FIG. 1, or by a direct wired connection, such as using a universal serial bus (USB) connection or like coupling aspect. After the video streams are encoded, the switching/routing tool **404** direct the encoded streams through applicable network infrastructure and/or other hardware to deliver the encoded streams to the conferencing software **406**. The conferencing software **406** transmits the encoded video streams to each connected client, such as the clients **408** and **410**, which receive and decode the encoded video streams to output the video content thereof for display by video output components of the clients, such as within respective user interface tiles of a user interface of the conferencing software **406**.

[0067] A user of the phone **412** participates in a conference using an audio-only connection and may be referred to an audio-only caller. To participate in the conference from the phone **412**, an audio signal from the phone **412** is received and processed at a VOIP gateway **414** to prepare a digital telephony signal for processing at the conferencing system **400**. The VOIP gateway **414** may be part of the system **100**, for example, implemented at or in connection with a server of the datacenter **106**, such as the telephony server **112** shown in FIG. 1. Alternatively, the VOIP gateway **414** may be located on the user-side, such as in a same location as the phone **412**. The digital telephony signal is a packet switched signal transmitted to the switching/routing tool **404** for delivery to the conferencing software **406**. The conferencing software **406** outputs an audio signal representing a combined audio capture for each participant of the conference for output by an audio output component of the phone **412**. In some implementations, the VOIP gateway **414** may be omitted, for example, where the phone **412** is a VOIP-enabled phone.

[0068] A conference implemented using the conferencing software **406** may be referred to as a video conference in which video streaming is enabled for the conference participants thereof. The enabling of video streaming for a conference participant of a video conference does not require that the conference participant activate or otherwise use video functionality for participating in the video conference. For example, a conference may still be a video conference where none of the participants joining using clients turns on their video stream for any portion of the conference. In some cases, however, the conference may have video disabled, such as where each participant connects to the conference using a phone rather than a client, or where a host of the conference selectively configures the conference to exclude video functionality.

[0069] The conferencing system **400** may be used to implement a video conference in any of a variety of possible use cases. For example, the conferencing system **400** can implement a participant-to-participant video conference in which participants as end users of the conferencing software **406** may have their own user interface tiles and video streams, connect from their own participant devices (e.g., the client **408** or the client **410**), have similar meeting controls, be or be granted host privileges, or the like. A participant-to-participant video conference as referred to herein may thus be recognized as a conventional video conference between two or more given people. In another example, the conferencing system **400** can implement a contact center engagement video conference in which one participant to the video conference is a contact center user who is accessing a contact center over a video modality facilitated wholly or partially using the conferencing software **406** and another participant to the video conference is a contact center agent who works with or otherwise for the contact center to address queries from contact center users. In some such cases, the conferencing system **400** may be implemented within a contact

center service to deliver real-time communications over a video modality of the contact center. In yet another example, the conferencing system **400** can implement an online learning video conference in which multiple participants are recognized generally as audience participants and one or more select participants are recognized generally as leader participants. An online learning video conference may, for example, be used for educational purposes (e.g., virtual classes for online-capable schools), webinar purposes, or other lecture- or presentation-type purposes. Other examples of video conferences implementable using the conferencing system **400** are possible.

[0070] In some implementations, other software services may be accessible in connection with a conference implemented using the conferencing system **400**. For example, a conference may include or otherwise integrate functionality for instant messaging, unified messaging, and other types of messaging communications between participants of the conference, such as to facilitate a chat or like virtual conversation between users of those participants. Those other software services may be implemented at the conferencing system **400** and/or a different aspect of the system **100**.

[0071] FIG. 5 is a block diagram of an example of an interactive video layer system **500**. The system **500** includes a server device **502**, a first participant device **504**, and a second participant device **506**. Each of the first participant device **504** and the second participant device **506** may, for example, be one of the clients **408** or **410** shown in FIG. 4. The first participant device **504** and the second participant device **506** are devices used by conference participants to connect to and participate in a video conference implemented by conferencing software **508** at the server device **502**. The conferencing software **508** may, for example, be the conferencing software **406** shown in FIG. 4. While two participant devices **504** and **506** are shown as connecting to a video conference implemented by the conferencing software **508** in a non-limiting example, in some implementations, other numbers of participant devices may be so connected.

[0072] The first participant device **504** connects to the conferencing software **508** using a client application **510** and the second participant device **506** connects to the conferencing software **508** using a client application **512**. In particular, the client application **510** and the client application **512** are client-side software applications used to connect the first participant device **504** and the second participant device **506**, respectively, to the video conference implemented by the conferencing software **508**. For example, the client application **510** and the client application **512** may be different instances, installations, or versions of a same software application, in which one is present at the first participant device **504** and the other is present at the second participant device **506**. In another example, the client application **510** and the client application **512** may be different software applications altogether. The client application **510** and/or the client application **512** may, for example, be a desktop software application, mobile application, or web application associated with one or more services of a software platform, for example, the software platform **300** shown in FIG. 3. For example, the client application **510** or the client application **512** may be software that allows a user of the first participant device **504** or the second participant device **506**, respectively, to access or otherwise use one or more of the software **312** through **318** shown in FIG. 3. In some implementations, the client application **510** and/or the client application **512** may be a web-based application accessible via a web browser of the first participant device **504** and/or the second participant device **506**, respectively.

[0073] Each of the first and second participant devices **504** and **506** includes components used for or otherwise during the video conference. In particular, the first participant device **504** includes a camera **514** and a display **516**, and the second participant device **506** includes an input component **518** and a display **520**. The display **516** and the display **520** are computer displays configured to output content of the video conference for viewing by users of the first participant device **504** and the second participant device **506**, respectively. For example, the display **516** and/or the display **520** may be an integrated (e.g., built-in) display of the first participant device **504** or the second participant device **506**, respectively, or may connect to the first participant device **504** or the second participant device **506** using a port, such as via a USB or a high definition multimedia interface

(HDMI) connection. The camera **514** is a component configured to capture a video stream at the first participant device **504**, which video stream will be transmitted to the conferencing software **508** for display to the various participants of the video conference during the video conference. For example, the camera **514** may be an integrated (e.g., built-in) camera of the first participant device **504** or may connect to the first participant device **504** using a port, such as via a USB connection. The input component **518** is a component configured to capture input from a user of the second participant device **506**, which input may be processed by the client application **512** in connection with the video conference. For example, the input component **518** may be a keyboard, a mouse, or a touchscreen (e.g., the display **520** itself, as applicable). While the first participant device **504** is not shown as including an input component **518** and the second participant device **506** is not shown as including a camera, in at least some cases, the first participant device **504** will also include an input component and/or the second participant device **506** will also include a camera.

[0074] The conferencing software **508** includes or otherwise uses interactive video layer software **522** to enable interactions with objects within video layers of video streams displayed during a video conference implemented using the conferencing software **508**. In particular, the interactive video layer software **522** enables one or more objects located in any of multiple video layers of a video stream a participant device connected to the video conference to be interacted with from other participant devices connected to the video conference.

[0075] The interactive video layer software **522** will in particular be described with respect to FIG. 5 by non-limiting example as enabling interactions with a video stream of the first participant device **504**, captured by the camera **514** and processed using the client application **510**, from the second participant device **506**, using the input component **518**. The interactive video layer software **522** processes the video stream from the first participant device **504** to identify multiple video layers of that video stream, including video layers initially included in the video stream when same is obtained at the server device **502** (e.g., by the conferencing software **508**) from the first participant device **504** and video layers added to the video stream by the interactive video layer software **522** to correspond to those initial video layers. The interactive video layer software **522** detects an interaction from the second participant device **506** within an object within one of the multiple video layers identified for the video stream from the first participant device **504**. Based on that interaction, the interactive video layer software **522** determines (e.g., identifies, generates, retrieves, computes, or accesses) information associated with the object and presents that information within a GUI associated with the video conference implemented by the conferencing software **508**. For example, the GUI within which the information associated with the object is presented may be a GUI which includes a user interface tile within which the video stream from the first participant device **504** is displayed. In particular, the information associated with the object may be presented within that user interface tile during the video conference, for example, as an overlay, annotation, or other content element alongside the video stream itself.

[0076] While the interactive video layer software **522** is shown and described as being included within the conferencing software **508**, in some implementations, the interactive video layer software **522** or a portion thereof may be external to the conferencing software **508**. For example, the interactive video layer software **522** may in some cases represent functionality of a software platform which includes the conferencing software **508** (e.g., the software platform **300**) external to but otherwise for use with the conferencing software **508**. In another example, the interactive video layer software **522** may be implemented at a participant device, such as within the client application **510** and/or the client application **512**. In one example where the interactive video layer software **522** is implemented at a participant device (e.g., within a client application running at that device or otherwise), the user of that participant device may cause their video stream to be interactive by enabling interactions with their video stream generally, such as via a user interface element (e.g., a meeting control) of the video conference, or by asserting configurations indicating one or more objects with which to enable or limit interactions from other participant devices. In

another example where the interactive video layer software **522** is implemented at a participant device, the user of that participant device may cause the video stream from another participant device to become interactive using a user interface element of the video conference or by performing an interaction with that video stream (e.g., clicking on a portion thereof corresponding to an object using a mouse of the participant device). In some such cases, a notification may be transmitted to the participant device from which the interacted video stream derived to inform a user thereof that their video stream is being interacted with. In some such cases, the notification may identify the interacted object or otherwise the portion of the video stream within which the interaction occurred.

[0077] The interactive video layer software **522** will be further described with respect to FIG. **6**, which shows a block diagram of an example of functionality of the interactive video layer software **522**. The interactive video layer software **522** includes tools, such as programs, subprograms, functions, routines, subroutines, operations, and/or the like, for enabling interactions with objects within video layers of video streams displayed during a video conference. As shown, the interactive video layer software **522** includes an interactive video layer generation tool **600**, an interaction configuration assertion tool **602**, an object interaction detection tool **604**, an object recognition tool **606**, and an object information processing tool **608**.

[0078] The interactive video layer generation tool **600** generates interactive video layers for a subject video stream (e.g., the video stream obtained from the first participant device **504**). Each interactive video layer corresponds to one video layer initially included in the video stream when that video stream is obtained (e.g., by the conferencing software **508**). The interactive video layer generation tool **600** may first determine the initial video layers of the video stream. Determining the initial video layers of the video stream may include performing segmentation to segment contents of the video stream into different layers. For example, determining the initial video layers of the video stream may include obtaining metadata or other data associated with the video stream from the conferencing software **508**. In one particular example, the conferencing software **508** may receive such metadata or other data within or alongside the video stream from the first participant device **504** (e.g., as side information transmitted along with or within the video stream). The metadata or other data identifies the video layers initially included in the video stream. For example, the metadata or other data may specifically identify the video layers specifically or indicate those video layers based on an identification of various video stream content as corresponding to those video layers. In another example, where the video stream is initially obtained as including a single video layer, the metadata or other data may identify or otherwise correspond to content of the video stream, and the interactive video layer generation tool **600** may separate the content the video stream into multiple video layers for further processing by the interactive video layer software **522** based on the metadata or other data.

[0079] To illustrate, the video stream obtained from the first participant device **504** may include a foreground layer and a background layer. The foreground layer includes content depicting the user of the first participant device **504**, as a participant to the subject video conference, and any objects which may be held by the user of the first participant device **504** at a given time during the video conference. The background layer includes content depicting what is behind the user of the first participant device **504** within the location at which the first participant device **504** is transmitting the video stream. For example, the background layer may include various objects, structures, or the like in a room in which the user of the first participant device **504** is seated during a video conference. The interactive video layer software **522** processes metadata or other data of the video stream itself or accompanying the video stream to identify the foreground layer and the background layer. For example, the metadata or other data may be metadata or data generated at the first participant device **504** to identify the separate foreground and background layers of the video stream. In another example, the metadata or other data identifies content of the video stream without separate video layers having been represented therein (e.g., by the client application **510**),

and the interactive video layer generation tool **600** processes such content to identify which corresponds to a foreground layer (i.e., based the content depicting a person or one or more objects held by the person) and which corresponds to a background layer (i.e., content not identified as corresponding to the foreground layer).

[0080] In some cases, the background layer may be or otherwise correspond to a virtual background of the conference participant. In such a case, the video stream is transmitted from the first participant device **504** as a composite video stream in which each video frame thereof is produced at the client application **510** by combining foreground content of the video stream (i.e., portions of the video stream depicting the user of the first participant device **504** and any objects held by them) with a virtual background selected by or otherwise for the user of the first participant device **504**. Where a virtual background is used, the metadata or other data obtained from the first participant device **504** indicates the use of the virtual background, and the virtual background signaled within the video stream will be identified as the background layer, with other content being identified as corresponding to one or more other initial video layers.

[0081] Once the initial video layers of the video stream are identified, interactive video layers corresponding to those initial video layers are generated and introduced within the video stream. Each interactive video layer corresponds to a different one of the initial video layers. Each of the initial video layers will have a single interactive video layer corresponding thereto. Referring to FIG. 7, an illustration of an example of video layers of a video stream **700** output for display during a video conference is shown. In particular, the video stream **700** is a video stream of a video conference participant, for example, the video stream obtained at the server device **502** from the first participant device **504**. The video stream **700** includes three layers initially (i.e., before interactive video layers are introduced)—a background layer **702**, a foreground layer **704**, and an overlay layer **706**.

[0082] The background layer **702** may, for example, depict objects behind the user of the first participant device **504** relative to a direction that the camera **514** is pointing. Alternatively, the background layer **702** may depict objects within a virtual background used to produce the video stream at the client application **510**. The foreground layer **704** may, for example, depict the user of the first participant device **504**. The overlay layer **706** may, for example, represent so-called “lower thirds” information such as information associated with the user of the first participant device **504** (e.g., name, pronunciation information, set of pronouns, or job title), information associated with information associated with an entity to which that user belongs (e.g., a name of or website for their corporate employer), or regional information (e.g., weather or location).

[0083] The interactive video layer generation tool **600** identifies the background layer **702**, the foreground layer **704**, and the overlay layer **706** and accordingly generates interactive video layers **708**, **710**, and **712**. In particular, the interactive video layer **708** is generated and introduced within the video stream **700** to correspond to the background layer **702**, the interactive video layer **710** is generated and introduced within the video stream **700** to correspond to the foreground layer **704**, and the interactive video layer **712** is generated and introduced within the video stream **700** to correspond to the overlay layer **706**. Generating the interactive video layers **708**, **710**, and **712** includes generating (e.g., producing or computing), for each of the respective layers **702**, **704**, and **706**, new video layers with content that is spatially mapped to co-located content in the respective layers **702**, **704**, and **706**. Introducing the interactive video layers **708**, **710**, and **712** includes adding the interactive video layers **708**, **710**, and **712** at appropriate locations within a stack of video layers for the video stream. For example, referring to the background layer **702** as a first or lowest layer, the interactive video layer **708** is added to the video stream layer stack after or otherwise on top of the background layer **702**. The foreground layer **704** is after or otherwise on top of the interactive video layer **708**, and the interactive video layer **710** is after or otherwise on top of the foreground layer **704**. Finally, the overlay layer **706** is after or otherwise on top of the interactive video layer **710**, and the interactive video layer **712** is after or otherwise on top of the

overlay layer **706**.

[0084] The interactive video layers introduced within the video stream are configured to receive interactions from one or more participant devices connected to the video conference. That is, because the content of a given initial layer (e.g., one of the layers **702**, **704**, and **706**) covers only a portion of the entire user interface tile display area within which the subject video stream is output for display, different visible portions of a video stream displayed within a given user interface tile are recognized as corresponding to potentially different video layers. For example, for a given video frame of the video stream which initially includes the layers **702**, **704**, and **706** and has been modified to include the interactive video layers **708**, **710**, and **712**, a conference participant may interact with a first portion depicting a person by interacting with the interactive video layer **710** (corresponding to the foreground layer **704** in which the person is depicted) or a second portion depicting an object behind the person by interacting with the interactive video layer **708** (corresponding to the background layer **702** in which the object is depicted).

[0085] Referring back to FIG. **6**, the interaction configuration assertion tool **602** asserts an interaction configuration against one or more relevant objects during a subject video conference to limit (e.g., temporarily or entirely prevent) interactions with those one or more objects during that video conference. An interaction configuration may be defined based on input obtained from a user of a participant device prior to or during a video conference. In particular, the user of the participant device may specify one or more objects for which to limit interactions from other participant devices during one or more video conferences (e.g., for an in-progress video conference, a next upcoming video conference, the next N video conferences (in which N is an integer greater than or equal to one), or all future video conferences). For example, where the configuration is for an in-progress video conference, the user of the participant device may specify one or more objects by dragging their mouse or otherwise using a touchscreen interface to surround the one or more objects (e.g., individually or as a group), thereby creating a bounding box surrounding the one or more objects with which to limit interactions by other conference participants. In another example, where the configuration is to be applied for an in-progress video conference, the user of the participant device may click on an object within their video stream to indicate to disallow interactions with it by other conference participants. In yet another example, where the configuration is for an in-progress video conference or specified or unspecified future video conference, the user of the participant device may specify the one or more objects with which to limit interactions by other conference participants by inputting text indicating a name and/or a description of the one or more objects.

[0086] In some implementations, an interaction configuration may instead identify one or more objects with which to enable interactions from other participant devices during a video conference. For example, rather than objects being generally interactive, the system **500** may allow the user of the first participant device **504** to specify certain objects which may be interacted with during an in-progress video conference, a next upcoming video conference, the next N video conferences (in which N is an integer greater than or equal to one) or all future video conferences. For example, where the configuration is to be applied for an in-progress video conference, the user of the participant device may specify one or more objects by dragging their mouse or otherwise using a touchscreen interface to surround the one or more objects (e.g., individually or as a group), thereby creating a bounding box surrounding the one or more objects to make interactive for other participants. In another example, where the configuration is to be applied for an in-progress video conference, the user of the participant device may click on an object within their video stream to indicate to allow interactions with it by other conference participants. In yet another example, where the configuration is for an in-progress video conference or specified or unspecified future video conference, the user of the participant device may specify the one or more objects to make interactive by inputting text indicating a name and/or a description of the one or more objects. In some implementations in which the interaction configurations indicate objects for which to enable

interactions, the interaction configurations may be defined or otherwise generated by a client application running at a participant device from which the subject video stream is obtained (e.g., the client application **510**).

[0087] The object interaction detection tool **604** detects interactions from participant devices connected to a video conference with objects within a video layer identified by the interactive video layer tool **600**. In particular, when another participant device (e.g., the second participant device **506**) interacts with the video stream of a participant device (e.g., the first participant device **504**), the object interaction detection tool **604** detects that the interaction has occurred and records or otherwise identifies a location within the GUI (displaying the subject video stream) at which the interaction occurred. For example, the location within the GUI can be spatially mapped to a particular interactive video layer of the video stream based on the portion of the video stream corresponding to the location at which the interaction occurred. The content at the subject location, which is of the video layer corresponding to the spatially mapped interactive video layer, may be identified as the object with which the detected interaction corresponds, also referred to as the interacted object.

[0088] The object interaction detection tool **604** may in some cases limit interactions with a given object from participant devices. For example, interactions with an object may be limited based on an interaction configuration asserted by or otherwise for a user of a participant device from which a video stream which includes that object is obtained. In another example, interactions with an object may be limited based on the object being partially or wholly hidden from view during the video conference. That is, given that a person depicted within a video stream may move while the camera of their participant device remains fixed at a current location (e.g., where the participant device is a computer using a webcam to capture the video stream therefrom) and/or the camera or the participant device does not remain at a fixed location during the video conference (e.g., where the participant device is a mobile device and the participant is moving the participant device during the video conference), certain movements during the video conference may from time to time cause one or more objects previously visible within the video stream to no longer be visible therein. For example, a conference participant may walk in front of the object and remain at that new position, thereby blocking the object from view. In such a case, where an object has become partially or wholly blocked, the object interaction detection tool **604** restricts interactions with the object based on a spatial mapping of the location within the GUI at which the interaction occurred to the object being unavailable.

[0089] The object recognition tool **606** determines information associated with the object with which an interaction was detected by the object interaction detection tool **604**. The information associated with the object includes information usable to identify one or more of what the specific object is, what type of object the object is, or a source (e.g., manufacturer, author, or artist) of the object. The information associated with the object may be determined in one or more ways. In one example, manual user input obtained from the participant device corresponding to the video stream (e.g., the first participant device **504**) may specify the information associated with the object. For example, the user of the participant device may enter the manual user input within one or more text fields of a client application used to connect the participant device to the video conference (e.g., the client application **510**). In some such cases, the manual user input may be presented when interaction configurations identifying objects which may be interacted with are asserted, or otherwise in connection with such assertion. For example, the user of the first participant device **504** may, for each object indicated within the interaction configurations as being capable of interaction from other participant devices, provide separate input with the information associated with those objects.

[0090] In another example, the information associated with the object may be accessed, retrieved, identified, or otherwise determined in response to a first interaction with the object during the video conference. For example, the object recognition tool **606** may receive an indication from the object

interaction detection tool **604** that a given object has been interacted with from a participant device. The object recognition tool **606** may thereafter transmit a request for identifying information of the object, such as to an external source **524**. The external source **524** is an information source external to the conferencing software **508** and the interactive video layer software **522** which may process a request received from the interactive video layer software **522** to determine and transmit, to the interactive video layer software **522**, the information associated with the object.

[0091] In one example, the external source **524** may be a text-based or image-based Internet search engine. For example, upon an object being identified as an interacted object, a screen capture depicting the object (e.g., cropped from within the video stream displayed within the GUI of the video conference) may be used for a reverse image search. In some such cases, the interactive video layer software **522** may invoke a call to an application programming interface (API) associated with the external source **524** to facilitate the reverse image search. In another example, upon the object being identified as an interacted object, text depicted on the object may be identified, such as using optical character recognition (OCR) processing, and used as the query for a text-based Internet search.

[0092] In another example, the external source **524** may be a specific website, software platform, software service, or the like which has been linked to the video conference, the conferencing software **508**, or the software platform implementing the conferencing software **508** (e.g., the software platform **300**). For example, the user of the first participant device **504**, an administrator of an account for an entity with which the user is associated, or another person may specify one or more such external sources as default external sources to be searched based on objects with which interactions are detected during a video conference. The default external sources may, for example, be or include online marketplaces, online encyclopedias, or websites associated with the providers of various goods and/or services. In some such cases, the interactive video layer software **522** may leverage an API of such a default external source to transmit the request for the information associated with the object thereto.

[0093] Once the information associated with the object has been determined, it may be stored in connection with the video conference for future use during the video conference, such as in response to a later interaction detected from the same participant device (e.g., the second participant device) or another participant device connected to the video conference. For example, storage of the information associated with the object may prevent unnecessary resource expenditure and latencies by recalling the information associated with the object at one or more other times during the video conference. Alternatively, in some cases, information associated with a given object may be determined as described above each time the object is interacted with during the video conference.

[0094] The object information processing tool **608** presents the information associated with the interacted object within a GUI associated with the video conference. The information associated with the object is presented within a GUI associated with the video conference. In particular, the information associated with the object is presented within a user interface tile within the GUI, in which the user interface tile is associated with the conference participant whose video stream includes the interacted object, and is visible only at the participant device from which the object interaction was detected. For example, the information associated with the object may be presented within a user interface tile which displays the video stream that includes the object. In one such case, the information associated with the object may be presented within a pop-up or prompt at a location of the object within the user interface tile. The user interface file associated with the user of the first participant device **504**, which is the one that displays the video stream, is included in the GUI associated with the video conference, which is output at both of the display **516** and the display **520**. However, because the interaction with the object was detected as being from the second participant device **506**, the information associated with the object may only be output within the GUI (i.e., within the user interface tile) at the display **520** and not also at the display **516**.

Alternatively, the information associated with interacted object may be presented within the GUI of the video conference at multiple or even all participant devices connected to the video conference. In another example, the information associated with the object may be presented other than within a user interface tile which displays the video stream that includes the object. In some such cases, the information associated with the object may be presented within a prompt or pop-up window accessible from within the video conference, such as a secondary window associated with the GUI of the video conference.

[0095] Although the tools **600** through **608** are shown as separate tools, in some implementations, two or more of the tools **600** through **608** may be combined into a single tool. Although the tools **600** through **608** are shown as functionality of the interactive video layer software **522** as a single piece of software, in some implementations, some or all of the tools **600** through **608** may exist outside of the interactive video layer software **522**. Similarly, in some implementations, a software service using the interactive video layer software **522** (e.g., the conferencing software **508**) may exclude the interactive video layer software **522** while still including the some or all of tools **600** through **608** in some form elsewhere or otherwise make use of the tools **600** through **608** while some or all of the tools **600** through **608** are included in some form elsewhere.

[0096] Example use cases are now described with reference to examples of GUIs which may be output for display to a participant device during a video conference facilitated using an interactive video layer system, such as the system **500**. FIGS. **8A-B** are illustrations of example GUIs of a participant-to-participant video conference facilitated using interactive video layers. In FIG. **8A**, a user **800** of a participant device (e.g., the first participant device **504**) is shown within a GUI of the video conference (e.g., within a user interface tile within which a video stream obtained from that participant device is displayed) as sitting on a couch in front of some objects, including a vase **802**, a clock **804**, and a painting **806**. At some point during the video conference, a user of another participant device (e.g., the second participant device **506**) may become interested with the vase **802** and interact with it, such as by clicking on it using a mouse (e.g., as the input component **518**). In FIG. **8B**, a pop-up **810** is presented within the GUI of the video conference based on the interaction with the vase **802**. The pop-up **810** includes information associated with the vase **802**, in this case, a determined name of the vase **802** and hyperlinked website at which the vase **802** may be available to purchase. For example, the name of the vase **802** may be determined by a reverse image search performed using an Internet search engine (e.g., as the external source **524**) using a cropped portion of the video stream depicting the vase **802**. In another example, the name of the vase **802** may have been pre-populated, such as by the user **800**. The user of the other participant device may accordingly interact with the hyperlink within the pop-up **810** to cause a web browser running at or otherwise installed on that other participant device to connect to the hyperlinked website.

[0097] The interaction with the vase **802** and/or any other object within the video stream depicted in the GUI of the video conference may be limited at a given time during the video conference based on whether or not the vase **802** and/or such other object is obscured from view within the GUI at such a given time and/or based on a configuration asserted by or on behalf of the user **800**. For example, while not shown, at some point during the video conference the user **800** may stand up from their couch and walk in front of the vase **802**, entirely or substantially blocking it from view within the video stream. In such a case, the vase **802** would no longer be available for interaction until it returns to view within the video stream (i.e., by the user **800** moving away from their position in front of the vase **802**). In some such cases, where the user of the other participant device had already interacted with the vase **802** so as to cause the pop-up **810** to be presented within the GUI of the video conference, the pop-up **810** may remain within the GUI notwithstanding the user **800** blocking the vase **802** from view within the video stream. However, in other such cases, the pop-up **810** may be temporarily or otherwise removed from the GUI of the video conference. In another example, a configuration asserted by or otherwise on behalf of the

user **800** may limit (e.g., temporarily or entirely prevent) interactions with one or more objects depicted within the video stream of the user **800**, for example, the clock **804**. In such a case, interactions by the other participant with the clock **804** will not cause a pop-up like the pop-up **810** to be presented within the GUI of the video conference.

[0098] FIGS. **9A-B** are illustrations of example GUIs of a contact center engagement video conference facilitated using interactive video layers. In FIG. **9A**, a user **900** of a participant device (e.g., the first participant device **504**) is a contact center agent depicted within a video stream obtained from that participant device. The user **900** is in particular depicted on a foreground layer of the video stream, while a window is depicted on a background layer thereof. The video conference enables a contact center user, who is using another participant device connected to the video conference (e.g., the second participant device **506**) to engage with the user **900** over a video modality, regardless of whether participant device of the contact center user is transmitting its own video stream depicting the contact center user. During the video conference, the contact center user asks the user **900** about how to link an external account to their account with a software platform associated with the contact center (e.g., the software platform **300**), and the contact center user mentions that they have asked about this same topic in the past with a different contact center agent.

[0099] Based on such discussion by the contact center user, in FIG. **9B**, an interactive object **904** and an interactive object **906** appear within the video stream of the user **900**, for example, within an overlay layer of the video stream. The interactive object **904**, when interacted with by the contact center user, will present frequently asked questions on how to link an account within the GUI of the video conference (e.g., by expanding the interactive object **904** within the GUI). The interactive object **906**, when interacted with by the contact center user, will present a transcript of the prior contact center engagement during which the contact center user inquired about account linking within the GUI of the video conference (e.g., by expanding the interactive object **906** within the GUI). In some implementations, clicking on the interactive object **904** or the interactive object **906** may cause a web browser running at the participant device of the contact center user to connect a webpage including the subject text.

[0100] FIGS. **10A-B** are illustrations of example GUIs of an online learning video conference facilitated using interactive video layers. In FIG. **10A**, a user **1000** of a participant device (e.g., the first participant device **504**) is a teacher giving a lesson to a group of students connected to the video conference. In particular, the user **1000** is testing the students' knowledge of a topic, in this case, computer icons. Text **1002** appearing within the GUI of the video conference asks the students, as other participants of the video conference, to identify which of four objects **1004**, **1006**, **1008**, or **1010** depicted within the GUI is commonly used as a "save icon." Here, the depicting of the user **1000** may be within a foreground layer of the video stream obtained from the participant device of the user **1000**, and the text **1002** as well as the objects **1004**, **1006**, **1008**, and **1010** may be depicted within an overlay layer of the video stream. Each of the students, from their own participant devices connected to the video conference (e.g., as various second participant devices **506**), may interact with (e.g., click on, using a mouse as the input component **518**) one of the objects **1004**, **1006**, **1008**, and **1010** so as to vote for the one that they believe correctly answers the question presented within the text **1002**.

[0101] In FIG. **10B**, the correct answer of the question presented within the text **1002** of FIG. **10A** is shown by text **1012**. The four objects **1004**, **1006**, **1008**, and **1010** previously shown in FIG. **10A** are replaced by a new object **1014** which includes the object **1004**, as the object which correctly answered the question presented within the text **1002**, surrounded by a border so as to draw attention thereto. The text **1012** identifies that the object **1004** was the correct answer. In this example, the text **1012** also indicates that **85** percent of votes from students were for the object **1004**. In some cases, a vote percentage may not be indicated within the text **1012**. In some implementations, the object **1004** may remain within GUI instead of the new object **1014** replacing

it. For example, the object **1004** alone may remain within the GUI so as to indicate that it is the correct answer to the question presented in the text **1002**, while the other objects **1006**, **1008**, and **1010** are removed from the GUI. In another example, all four of the objects **1004**, **1006**, **1008**, and **1010** may remain within the GUI. In some cases where all four of the objects **1004**, **1006**, **1008**, and **1010**, the object **1004** may be visually emphasized, for example, by a box being presented around it, by an appearance of the other objects **1006**, **1008**, and **1010** being changed (e.g., by those objects **1006**, **1008**, and **1010** appearing faded), or the like.

[0102] Other use cases beyond those described above with respect to FIGS. **8A-10B** may be used with an interactive video system as disclosed herein, such as the system **500**. For example, in a participant-to-participant video conference example, a book shown in a user's background can be interacted with to deliver the interacting user to an online marketplace page at which they can buy the book. In such a case, the interactive video layer system may, for example, perform object recognition to determine that the object is a book and then search an online book marketplace based on text identified on the object (e.g., an author and/or title of the book). In another participant-to-participant video conference example, a conference participant who experiences poor connection quality (e.g., due to local bandwidth limitations) may be presented with an interactive object within their own user interface tile or that of another conference participant, which when interacted with, indicates such connectivity issues to the other conference participants and optionally connects the participant with a website with common troubleshooting tips. In yet another participant-to-participant video conference example, a uniform resource locator (URL) appearing within shared media content shared during the video conference may be recognized and made into an interactive object that, when interacted with, will cause a web browser at the participant device of a subject conference participant to open and navigate to the URL. In a contact center engagement video conference example, the contact center agent may have assigned to them a virtual background populated using to include objects representing or linking content relevant to the subject contact center user, such as their account number, knowledgebase articles, topics for troubleshooting, past tickets, or the like. In a webinar video conference example, a featured product can be interacted with in the video conference to open a link to purchase the product, a document with marketing materials for the product, or the like. In an online learning video conference example or an entertainment-based video conference example (e.g., in which a television show or like visual publication is presented within a GUI of the video conference), a poll may track interactions by conference participants with various objects, such as to allow audience members of a reality television show to vote for their favorite characters by clicking on them within a video stream of the television show, or to allow students remotely connecting to an online lecture to vote for their favorite books by clicking on them within such a video stream. In another online learning video conference example or an entertainment-based video conference example, audience members may be given a task to locate and interact with a certain object or type of object, and the first to do so may be awarded a prize (e.g., extra credit for a student or a shout out for a television show audience member). In a general example applicable to one or more types of video conference, a captcha-type system may be implemented to authenticate one or more participants to the video conference by requiring those one or more participants to interact with certain objects or certain types of objects within a video stream. For example, access to the video conference by a participant may be limited based on a successful response to such an authentication request.

[0103] In some implementations, an interactive video system, such as the system **500**, may be used to deliver interactive video layer functionality as disclosed herein for media other than video streams obtained from participant devices connected to a video conference. For example, a video conference implemented by the conferencing software **508** may be a virtual conference within which users of participant devices are visually represented using graphical models, such as avatars. In such a case, the contents of the video stream from a participant device may be entirely computer-generated rather than captured from the real world using a camera. Initial video layers may be

identified based on the segmentation operations described above, and interactive video layers may be generated and introduced within the video stream based on the interactive video layer operations described above.

[0104] In some implementations, an interactive video system, such as the system **500**, may be used to deliver interactive video layer functionality as disclosed herein for three-dimensional video rather than conventional two-dimensional video as represented using the video streams described above. For example, the system **500** may build (e.g., determine, identify, or otherwise develop) an understanding of locations of contents of a three-dimensional video stream so as to determine the visibility of certain objects from certain vantages. Thus, whereas an object is either visible or not within a conventional two-dimensional video stream based on the video stream being according to the contents captured by a single camera, an object may be visible or not at different locations within a three-dimensional video stream. As such, in some such implementations, multiple perspectives of a given view within the three-dimensional video stream can each be identified as a different video layer.

[0105] In some implementations, an interactive video system, such as the system **500**, may collect information associated with interactions with objects from participant devices within one or more video conferences for user experience testing. For example, locations, instances, times, and other information associated with interactions with objects may be collected over time and analyzed to determine where conference participants most often interact with objects within a video stream (e.g., the upper right corner), objects that are most frequently interacted with (e.g., books), times during a video conference at which objects are most frequently interacted with (e.g., the beginning, middle, or end), or the like. The collected data may, for example, be used to refine the interactive video system, such as by arranging a GUI of the conferencing software **508** to include certain types of objects, objects at certain video stream locations, and/or objects at certain times during a video conference.

[0106] To further describe some implementations in greater detail, reference is next made to examples of techniques which may be performed by or using an interactive video layer system.

FIG. **11** is a flowchart of an example of a technique **1100** for enabling and processing interactions with objects within video layers of a video conference. The technique **1100** can be executed using computing devices, such as the systems, hardware, and software described with respect to FIGS. **1-10B**. The technique **1100** can be performed, for example, by executing a machine-readable program or other computer-executable instructions, such as routines, instructions, programs, or other code. The steps, or operations, of the technique **1100**, or another technique, method, process, or algorithm described in connection with the implementations disclosed herein can be implemented directly in hardware, firmware, software executed by hardware, circuitry, or a combination thereof.

[0107] For simplicity of explanation, the technique **1100** is depicted and described herein as a series of steps or operations. However, the steps or operations of the technique **1100** in accordance with this disclosure can occur in various orders and/or concurrently. Additionally, other steps or operations not presented and described herein may be used. Furthermore, not all illustrated steps or operations may be required to implement a technique in accordance with the disclosed subject matter.

[0108] At **1102**, multiple video layers of a video stream obtained from a first participant device connected to a video conference are identified during the video conference. The multiple video layers include multiple initial video layers and multiple interactive video layers corresponding to ones of those initial video layers. The initial layers may include various types of layers, for example, two or more of a background layer depicting background content of the video stream, a foreground layer depicting one or more participants of the video conference, and an overlay layer depicting virtual content of the video stream. Each interactive video layer corresponds to a different one of the initial video layers. For example, the multiple video layers may include at least a foreground layer and a background layer, in which a first interactive video layer is introduced to

enable interactions with the foreground layer and a second interactive video layer is introduced to enable interactions with the background layer. The multiple interactive video layers may be generated and introduced within the video stream.

[0109] At **1104**, an interaction with an object within a video layer of the multiple video layers from a second participant device connected to the video conference is detected during the video conference. The interaction is detected within one of the multiple interactive video layers introduced within the video stream. The interaction may have a different purpose and/or result based on a type of the video conference, a type of the interacted object, or the like. In some cases, the interaction with the object may be based on a request to a user of the participant device from which the interaction is detected. For example, the interaction may correspond to a selection of the object during a voting activity (e.g., a poll, quiz, or the like). In another example, the interaction may correspond to an object presented within a video stream of a contact center agent for interaction by a contact center user in an active contact center engagement with that contact center agent. In some cases, the interaction with the object may be initiated by user action at a participant device, such as without a request from the participant device from which the video stream is obtained. For example, the interaction may correspond to an object within a background of a participant-to-participant video conference in which the user of a participant device wants to learn more about the object.

[0110] At **1106**, information associated with the object is determined during the video conference. Determining the information associated with the object may include performing object recognition against content of the video stream at the a location within the one of the multiple video layers at which the interaction occurred. Alternatively, determining the information associated with the object may include obtaining input describing the object, such as from a participant device from which the video stream is obtained. The information associated with the object may be based on the type of the object and/or the type of the video conference. For example, where the interaction with the object corresponds to a selection of the object during a voting activity, the information associated with the object may indicate a vote for the object. In another example, the information associated with the object may include a URL associated with the object (e.g., for a website at which the object may be purchased), and determining the information associated with the object may accordingly include determining the URL based on the object (e.g., via a reverse image search or a text-based search). In some cases, where the information associated with the object has already been determined during the video conference, determining the information associated with the object based on the detected interaction therewith may include accessing a record or other data stored in connection with the video conference to include the information associated with the object.

[0111] At **1108**, the information associated with the object is presented within a GUI of the video conference during the video conference. For example, the information associated with the object may be presented within a user interface tile associated with the participant from whose participant device the video stream is obtained. In some such cases, the information associated with the object may be presented within a pop-up within that user interface tile. In another example, the information associated with the object may be presented within a pop-up, prompt, or other window of the GUI of the video conference. In some cases, the information associated with the object may itself include an interactive element that, when interacted with from a participant device, may trigger some action. For example, the information associated with the object may include a URL for a website at which an interacted object may be purchased, and a user of a participant device from which the object interaction is detected may further interact with that URL within a pop-up in the GUI of the video conference to cause a web browser at their participant device to be directed to the subject website.

[0112] In some implementations, the technique **1100** may include asserting interaction configurations against one or more objects shown in the video stream to control interactions with

the one or more objects during the video conference. For example, a user of the participant device from which the video stream is obtained may assert the interaction configurations to prevent one or more objects from being interacted with during the video conference. In another example, the user of the participant device from which the video stream is obtained may assert the interaction configurations to enable interactions with one or more objects within the video stream. The interaction configurations may, for example, be defined at the participant device from which the video stream is obtained.

[0113] In some implementations, the technique **1100** may include limiting interactions with the object while the object is hidden from view during the video conference. For example, the interaction detected above may be a first interaction with the object. After that first interaction, the user of the participant device from which the video stream is obtained may move in front of the object so as to partially or wholly block it from view within the video stream. In such a case, a further interaction with that object may be precluded while the object remains blocked from view. For example, further interactions with that object may be re-enabled upon the user of the participant device moving away from the object such that the object becomes visible once again.

[0114] In some implementations, the technique **1100** may include signaling data associated with the interaction with the object to a host of the video conference. For example, the host of the video conference may receive information identifying interactions with video stream objects during the video conference from participant devices at one or more times during the video conference or after the video conference ends. The information may, for example, participants who interacted with objects, participants whose video streams were interacted with, and/or objects which were interacted with, along with numbers and/or times of occurrences of such interactions. In some such implementations, the data may be signaled to the host of the video conference within a summary of the video conference after the video conference ends, such as along with a copy of a recording and/or transcript of the video conference.

[0115] The implementations of this disclosure correspond to methods, non-transitory computer readable media, apparatuses, systems, devices, and the like. In some implementations, a method comprises identifying, during a video conference, multiple video layers of a video stream obtained from a first participant device connected to the video conference; detecting, during the video conference, an interaction with an object within one of the multiple video layers from a second participant device connected to the video conference; and presenting, during the video conference, information associated with the object within a graphical user interface associated with the video conference based on the interaction. In some implementations, a non-transitory computer readable medium stores instructions operable to cause one or more processors to perform operations comprising identifying, during a video conference, multiple video layers of a video stream obtained from a first participant device connected to the video conference; detecting, during the video conference, an interaction with an object within one of the multiple video layers from a second participant device connected to the video conference; and presenting, during the video conference, information associated with the object within a graphical user interface associated with the video conference based on the interaction. In some implementations, an apparatus comprises a memory and a processor configured to execute instructions stored in the memory to identify, during a video conference, multiple video layers of a video stream obtained from a first participant device connected to the video conference; detect, during the video conference, an interaction with an object within one of the multiple video layers from a second participant device connected to the video conference; and present, during the video conference, information associated with the object within a graphical user interface associated with the video conference based on the interaction.

[0116] In some implementations of the method, non-transitory computer readable medium, or apparatus, the method comprises, the operations comprise, and the processor is configured to execute the instructions for introducing multiple interactive video layers to the video stream, wherein each interactive video layer of the multiple interactive video layers corresponds to a

different one of the multiple video layers, and wherein the interaction is detected within one of the multiple interactive video layers.

[0117] In some implementations of the method, non-transitory computer readable medium, or apparatus, the method comprises, the operations comprise, and the processor is configured to execute the instructions for asserting interaction configurations against one or more objects shown in the video stream to control interactions with the one or more objects during the video conference.

[0118] In some implementations of the method, non-transitory computer readable medium, or apparatus, the interaction corresponds to a selection of the object during a voting activity, and the information associated with the object indicates a vote for the object.

[0119] In some implementations of the method, non-transitory computer readable medium, or apparatus, the information associated with the object includes a uniform resource locator associated with the object, and the method comprises, the operations comprise, and the processor is configured to execute the instructions for determining the uniform resource locator based on the object.

[0120] In some implementations of the method, non-transitory computer readable medium, or apparatus, the method comprises, the operations comprise, and the processor is configured to execute the instructions for limiting interactions with the object while the object is hidden from view during the video conference.

[0121] In some implementations of the method, non-transitory computer readable medium, or apparatus, the method comprises, the operations comprise, and the processor is configured to execute the instructions for signaling data associated with the interaction to a host of the video conference.

[0122] In some implementations of the method, non-transitory computer readable medium, or apparatus, the video conference is between a contact center user and a contact center agent, the video stream is of the contact center agent, and the object is presented for interaction by the contact center user.

[0123] In some implementations of the method, non-transitory computer readable medium, or apparatus, the video conference is a virtual conference in which participants are visually represented by avatars and the video stream depicts virtual content associated with one of the avatars.

[0124] In some implementations of the method, non-transitory computer readable medium, or apparatus, the multiple video layers include two or more of a background layer depicting background content of the video stream, a foreground layer depicting one or more participants of the video conference, and an overlay layer depicting virtual content of the video stream.

[0125] In some implementations of the method, non-transitory computer readable medium, or apparatus, the method comprises, the operations comprise, and the processor is configured to execute the instructions for determining the information associated with the object by performing object recognition against content of the video stream at the a location within the one of the multiple video layers at which the interaction occurred.

[0126] In some implementations of the method, non-transitory computer readable medium, or apparatus, identifying the multiple video layers comprises identifying multiple initial video layers of the video stream as received from the first participant device; and generating a different interactive video layer for each of the multiple initial video layers, wherein the multiple video layers includes the multiple initial video layers and each of the different interactive video layers.

[0127] In some implementations of the method, non-transitory computer readable medium, or apparatus, the interaction with the object is based on a request to a user of the second participant device.

[0128] In some implementations of the method, non-transitory computer readable medium, or apparatus, the multiple video layers include at least a foreground layer and a background layer, and

a first interactive video layer is introduced to enable interactions with the foreground layer and a second interactive video layer is introduced to enable interactions with the background layer.

[0129] In some implementations of the method, non-transitory computer readable medium, or apparatus, the method comprises, the operations comprise, and the processor is configured to execute the instructions for limiting interactions with the object based on a configuration asserted for a user of the first participant device.

[0130] In some implementations of the method, non-transitory computer readable medium, or apparatus, the information associated with the object includes a hyperlink to a website associated with the object.

[0131] In some implementations of the method, non-transitory computer readable medium, or apparatus, the object is introduced within the one of the multiple video layers based on a connection quality of the second participant device to the video conference.

[0132] In some implementations of the method, non-transitory computer readable medium, or apparatus, the object is introduced within the one of the multiple video layers based on a discussion during the video conference.

[0133] The implementations of this disclosure can be described in terms of functional block components and various processing operations. Such functional block components can be realized by a number of hardware or software components that perform the specified functions. For example, the disclosed implementations can employ various integrated circuit components (e.g., memory elements, processing elements, logic elements, look-up tables, and the like), which can carry out a variety of functions under the control of one or more microprocessors or other control devices. Similarly, where the elements of the disclosed implementations are implemented using software programming or software elements, the systems and techniques can be implemented with a programming or scripting language, such as C, C++, Java, JavaScript, assembler, or the like, with the various algorithms being implemented with a combination of data structures, objects, processes, routines, or other programming elements.

[0134] Functional aspects can be implemented in algorithms that execute on one or more processors. Furthermore, the implementations of the systems and techniques disclosed herein could employ a number of conventional techniques for electronics configuration, signal processing or control, data processing, and the like. The words “mechanism” and “component” are used broadly and are not limited to mechanical or physical implementations, but can include software routines in conjunction with processors, etc. Likewise, the terms “system” or “tool” as used herein and in the figures, but in any event based on their context, may be understood as corresponding to a functional unit implemented using software, hardware (e.g., an integrated circuit, such as an ASIC), or a combination of software and hardware. In certain contexts, such systems or mechanisms may be understood to be a processor-implemented software system or processor-implemented software mechanism that is part of or callable by an executable program, which may itself be wholly or partly composed of such linked systems or mechanisms.

[0135] Implementations or portions of implementations of the above disclosure can take the form of a computer program product accessible from, for example, a computer-usable or computer-readable medium. A computer-usable or computer-readable medium can be a device that can, for example, tangibly contain, store, communicate, or transport a program or data structure for use by or in connection with a processor. The medium can be, for example, an electronic, magnetic, optical, electromagnetic, or semiconductor device.

[0136] Other suitable mediums are also available. Such computer-usable or computer-readable media can be referred to as non-transitory memory or media, and can include volatile memory or non-volatile memory that can change over time. The quality of memory or media being non-transitory refers to such memory or media storing data for some period of time or otherwise based on device power or a device power cycle. A memory of an apparatus described herein, unless otherwise specified, does not have to be physically contained by the apparatus, but is one that can

be accessed remotely by the apparatus, and does not have to be contiguous with other memory that might be physically contained by the apparatus.

[0137] While the disclosure has been described in connection with certain implementations, it is to be understood that the disclosure is not to be limited to the disclosed implementations but, on the contrary, is intended to cover various modifications and equivalent arrangements included within the scope of the appended claims, which scope is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures as is permitted under the law.

## Claims

1. A method, comprising: detecting, during a contact center engagement video conference between a contact center user device and a contact center agent device, an interaction with an object depicted within one of multiple layers of a video stream of the contact center agent device based on input from the contact center user device; and presenting, based on the interaction, information associated with the object within a graphical user interface associated with the contact center engagement video conference.
2. The method of claim 1, comprising: obtaining the video stream from the contact center agent device; and introducing multiple interactive layers to the video stream, wherein each of the multiple interactive layers corresponds to a different one of the multiple layers.
3. The method of claim 1, comprising: controlling interactions by the contact center user device with objects depicted within the video stream during the contact center engagement video conference based on configurations asserted against the objects.
4. The method of claim 1, comprising: presenting the object within the video stream based on content of the contact center engagement video conference.
5. The method of claim 1, comprising: including data associated with the interaction within one or both of a recording of the contact center engagement video conference or a transcript of the contact center engagement video conference.
6. The method of claim 1, wherein the information is presented within a user interface tile which displays the video stream within the graphical user interface.
7. The method of claim 1, wherein the object corresponds to a contact center engagement transcript.
8. The method of claim 1, wherein the multiple layers of the video stream include two or more of a first interactive layer corresponding to background content of the video stream, a second interactive layer corresponding to foreground content of the video stream, or a third interactive layer corresponding to overlay content of the video stream.
9. The method of claim 1, wherein the object is one of multiple objects depicted within the video stream.
10. The method of claim 1, wherein the object is presented for interaction by the contact center user device during the contact center engagement video conference based on content of the contact center engagement video conference.
11. A non-transitory computer readable medium storing instructions operable to cause one or more processors to perform operations comprising: detecting, during a contact center engagement video conference between a contact center user device and a contact center agent device, an interaction with an object depicted within one of multiple layers of a video stream of the contact center agent device based on input from the contact center user device; and presenting, based on the interaction, information associated with the object within a graphical user interface associated with the contact center engagement video conference.
12. The non-transitory computer readable medium of claim 11, wherein the multiple layers include a foreground layer, a background layer, and an overlay layer and the operations comprise: introducing a first interactive layer for the foreground layer, a second interactive layer for the

- background layer, and a third interactive layer for the overlay layer, wherein the interaction with the object is in one of the first interactive layer, the second interactive layer, or the third interactive layer.
- 13.** The non-transitory computer readable medium of claim 11, wherein the interaction is controlled based on a configuration asserted against the object.
- 14.** The non-transitory computer readable medium of claim 11, wherein the interaction is based on a request to a user of the contact center user device.
- 15.** A system, comprising: one or more memories; and one or more processors configured to execute instructions stored in the one or more memories to: detect, during a contact center engagement video conference between a contact center user device and a contact center agent device, an interaction with an object depicted within one of multiple layers of a video stream of the contact center agent device based on input from the contact center user device; and present, based on the interaction, information associated with the object within a graphical user interface associated with the contact center engagement video conference.
- 16.** The system of claim 15, wherein the one or more processors are configured to execute the instructions to: introduce, for each layer of the multiple layers, an interactive layer configured to receive interactive input from the contact center user device.
- 17.** The system of claim 15, wherein the interaction is indicated within one or both of a recording of the contact center engagement video conference or a transcript of the contact center engagement video conference.
- 18.** The system of claim 15, wherein the one of the multiple layers is a foreground layer, a background layer, or an overlay layer.
- 19.** The system of claim 15, wherein the information is presented within a pop-up or prompt at a location of the object within a user interface tile of the graphical user interface.
- 20.** The system of claim 15, wherein the object corresponds to a knowledgebase article or frequently asked questions.
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